

Computer Architecture Lab

(EL-2012)

LABORATORY REPORT

Fall 2026



(Lab#3)

Apply Behavioral-level Verilog to Design and Simulate Combinational Modules

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MARKS AWARDED: _____ / 10

Signature and date of Instructor: _____

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Verified by: Dr. Nasir Ali Shah

Updated: Fall 2026

Task 1:

Description:

// Write description of the task here 🖱 (max. five to seven lines)

In this task , we I write behavioural level Verilog code for adder and subtractor . to add or subtract 4 bit number I use control flow statements. And also write its test bench and simulate the code and check the behaviour of wave form which matches my required output

Code:

// Paste source code here 🖱

```
module add_sub(a,b,c,ct);  
input [3:0]a;  
input [3:0]b;  
input ct;  
output reg [3:0]c;  
always @ *  
begin  
if (ct==0)  
c=a+b;  
else  
c=a-b;  
end  
endmodule
```

```
module add_sub_tb;  
reg [3:0]a;  
reg [3:0]b;  
reg ct;  
wire [3:0]c;  
add_sub uut(  
.a(a),  
.b(b),  
.ct(ct),  
.c(c)  
);  
initial  
begin
```

```
a=4'b0010;
b=4'b0001;
ct=0;
#50;
a=4'b0010;
b=4'b0001;
ct=1;
#50;
a=4'b1010;
b=4'b0001;
ct=0;
#50;
a=4'b1010;
b=4'b0001;
ct=1;
#50;
a=4'b1010;
b=4'b0011;
ct=0;
#50;
a=4'b1010;
b=4'b0011;
ct=1;
#50;
end
endmodule
```

Output:

// Import output here



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Task 2:

Description:

// Write description of the task here  (max. five to seven lines)

In this task , we I write behavioural level Verilog code for 2x1 multiplexer. to make 2x1 multiplexer I use control flow statements. And also write its test bench and simulate the code and check the behaviour of wave form which matches my required output

Code:

// Paste source code here 

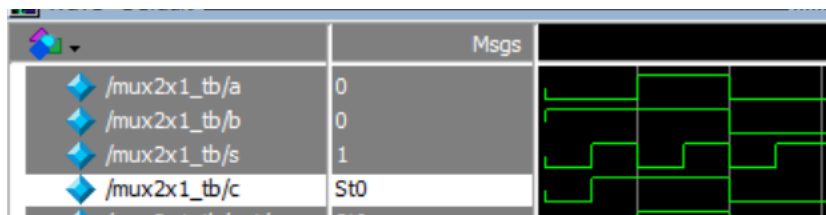
```
module mux2x1(a,b,c,s);  
input a,b,s;  
output reg c;  
always @*  
begin  
    if(s==0)  
        c=a&b;  
    else  
        c=a|b;  
end  
endmodule
```

```
module mux2x1_tb;  
reg a,b,s;  
wire c;  
mux2x1 uut(  
    .a(a),  
    .b(b),  
    .s(s),  
    .c(c)  
);  
initial  
begin
```

```
a=0;
b=1;
s=0;
#50;
a=0;
b=1;
s=1;
#50;
a=1;
b=1;
s=0;
#50;
a=1;
b=1;
s=1;
#50;
a=0;
b=0;
s=0;
#50;
a=0;
b=0;
s=1;
#50;
end
endmodule
```

Output:

// Import output here 



Task 3:

Description:

// Write description of the task here 🖱 (max. five to seven lines)

In this task , we I write behavioural level Verilog code for 2 bit comparator which tells us whether the 2 numbers are rqual, greater, or lesser. To make 2 bit comparator I use control flow statements. And also write its test bench and simulate the code and check the behaviour of wave form which matches my required output

Code:

// Paste source code here 🖱

```
module comparator(a,b,g,e,l);
input [1:0]a;
input [1:0]b;
output reg g;
output reg e;
output reg l;
always @*
begin
g=2'b00;
l=2'b00;
e=2'b00;
if(a>b)
g=1;
else if(a==b)
e=1;
else
l=1;
end
endmodule;
module comparator_tb;
reg [1:0]a;
reg [1:0]b;
wire g,e,l;
```

```

comparator uut(
.a(a),
.b(b),
.g(g),
.l(l),
.e(e)
);
initial
begin
a=2'b00;
b=2'b01;
#50;
a=2'b11;
b=2'b01;
#50;
a=2'b01;
b=2'b01;
#50;
a=2'b10;
b=2'b01;
#50;
end
endmodule

```

Output:

// Import output here 🖱️

